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BE 1st Year, 3rd Test Dec 2023
Applied Physics 2APRC2 (ETC B and E&I)

Time: 70 min

Max. Marks: 20

All questions are compulsory. Attempt any one part from each question.

- Q1 I (a)** A photon of frequency ν interacts with an electron at rest. Obtain an expression for change of wavelength in terms of scattered angle. Calculate the Compton shift if x-rays of $\lambda=1\text{\AA}$ are scattered from carbon block. The scattered radiation is viewed at 90° to the incident beam. 8
- (b)** What is matter wave and its properties? What voltage must be applied to an electron microscope to produce electrons of $\lambda=0.5\text{\AA}$. 5

OR

- II(a)** Derive the time dependent and time independent Schrödinger's wave equation for free particle. 7
- (b)** What is Heisenberg's uncertainty principle for time and Energy? Explain this with electron diffraction with single slit experiments? Calculate the minimum uncertainty in the velocity of an electron confined to a box of 10^{-8}m length. 6
- Q2 I(a)** What is lattice plane? Derive the formula for the distance between two adjacent planes of a simple cubic lattice. 5
- (b)** X-ray of wavelength $1.18\text{\AA}$ are diffracted by a plane (111) in a cubic crystal at an angle 60° in first order reflection. Calculate the interatomic distance. 2

OR

- II(a)** Explain the seven crystal structure and Bravais lattice? 5
- (b)** For an fcc cubic crystal, atomic radius of (110) plane is $2\sqrt{2}\text{\AA}$. Calculate interplanar spacing. 2
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